

Section 5 - Weather Report Automation

Project Example: Weather Report

I want to be able to select a location from a dropdown list and get the high and the low temperatures for today in that specific location I select.

After I get the high and low temperature I want to map those values to a database in Airtable so I see all of my past runs of this automation and track the weather data.



Automation Workflow Overview



1. Tool Stack

- **N8N**: No-code/low-code automation platform.
 - **Airtable**: Used as a flexible backend database.
 - **Open-Meteo API**: Free weather API that doesn't require an API key.
 - **JavaScript (via N8N code node)**: Used to process weather data.
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Step-by-Step Breakdown

◆ 1. Setting Up Airtable Database

- Create a new **base** called `Weather Center`.
 - Table fields:
 - `location`: Text field (e.g., "Chicago")
 - `high`: Number (Fahrenheit)
 - `low`: Number (Fahrenheit)
 - `created_time`: Auto-generated timestamp
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◆ 2. Create N8N Workflow

◆ Trigger: N8N Form

- Add a dropdown form asking **"Select your location"** with options: Chicago, New York, Miami, Phoenix.
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◆ 3. Routing Using Switch Node

- A **Switch node** is used to route based on the selected location.
 - Routes: Chicago, New York, Miami, Phoenix.
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◆ 4. Setting Coordinates

Each route sets **latitude and longitude** for that city using a **Set** node:

- **Chicago:** lat: 41.87 , lon: -87.63
- **New York:** lat: 40.71 , lon: -74.00
- **Miami:** lat: 25.76 , lon: -80.19
- **Phoenix:** lat: 33.45 , lon: -112.07

✓ **Note:** Be sure to include the **minus (-)** sign for longitude to avoid pulling incorrect locations.

◆ 5. HTTP Request to Open-Meteo

- Uses **GET** method with dynamic URL:

```
bash
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https://api.open-meteo.com/v1/forecast?latitude={{ $json.latitude }}&longitude={{ $json.longitude }}&hourly=temperature_2m&temperature_unit=fahrenheit
```

- Returns JSON weather data including hourly temperature.
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◆ 6. JavaScript Node (Code Node)

- Parse the JSON to extract:
 - **Highest temperature**
 - **Lowest temperature**
- The structure returned from Open-Meteo contains:

```
json
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{
  "hourly": {
    "temperature_2m": [71.6, 75.2, 78.8, ...]
  }
}
```

You'd use JavaScript like this:

```
javascript
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const temps = items[0].json.hourly.temperature_2m;
const high = Math.max(...temps);
const low = Math.min(...temps);

return [{
  json: {
    high,
    low,
    location: items[0].json.location,
    created_time: new Date().toISOString()
  }
}];
```

◆ 7. Send Parsed Data to Airtable

- Connect to Airtable via **Airtable node** in N8N.
- Use the **Create Record** action and map:
 - **location**
 - **high**
 - **low**
 - **created_time**